COMPARISON OF THE LIMIT BIG FPPF TOPOLOGIES WITH THE FPPF TOPOLOGY ON THE LIMIT SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [4508, 4541] and proof(s) [[4543, 4581]]; this spec covers the complete environment instead (statement 4508-4541, proof span(s) [(4543, 4581)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original definition 020R: sources/stacks-chow/tags.txt, source lines 2644–2644. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0219: sources/stacks-chow/tags.txt, source lines 2662–2662. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0223: sources/stacks-chow/tags.txt, source lines 2692–2692. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0229: sources/stacks-chow/tags.txt, source lines 2698–2698. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 021Q: sources/stacks-chow/tags.txt, source lines 2679–2679. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01ZN: sources/stacks-chow/tags.txt, source lines 2600–2600. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04AI: sources/stacks-chow/tags.txt, source lines 5617–5617. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original reference 0EUU: sources/stacks-chow/tags.txt, source lines 18564–18564. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07RP: sources/stacks-chow/tags.txt, source lines 9874–9874. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C0C: sources/stacks-chow/tags.txt, source lines 15082–15082. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0C3L: sources/stacks-chow/tags.txt, source lines 15196–15196. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Let T be an affine scheme which is written as a limit $T = \lim_{i \in I} T_i$ of a directed inverse system of affine schemes.*

1. *Let $\mathcal{V} = \{V_j \to T\}_{j=1,\ldots,m}$ be a standard τ-covering of T, see Definitions 020R, 0219, 0223, 0229, and 021Q. Then there exists an index i and a standard τ-covering $\mathcal{V}_i = \{V_{i,j} \to T_i\}_{j=1,\ldots,m}$ whose base change $T \times_{T_i} \mathcal{V}_i$ to T is isomorphic to $\mathcal{V}$.*
2. *Let $\mathcal{V}_i$, $\mathcal{V}'_i$ be a pair of standard τ-coverings of T_i. If $f : T \times_{T_i} \mathcal{V}_i \to T \times_{T_i} \mathcal{V}'_i$ is a morphism of coverings of T, then there exists an index $i' \geq i$ and a morphism $f_{i'} : T_{i'} \times_{T_i} \mathcal{V} \to T_{i'} \times_{T_i} \mathcal{V}'_i$ whose base change to T is f.*
3. *If $f, g : \mathcal{V} \to \mathcal{V}'_i$ are morphisms of standard τ-coverings of T_i whose base changes f_T, g_T to T are equal then there exists an index $i' \geq i$ such that $f_{T_{i'}} = g_{T_{i'}}$.*

In other words, the category of standard τ-coverings of T is the colimit over I of the categories of standard τ-coverings of T_i.

Proof. Let us prove this for $\tau = fppf$. By Limits, Lemma 01ZM the category of schemes of finite presentation over T is the colimit over I of the categories of finite presentation over T_i. By Limits, Lemmas 01ZN and 04AI the same is true for category of schemes which are affine, flat and of finite presentation over T. To finish the proof of the lemma it suffices to show that if $\{V_{j,i} \to T_i\}_{j=1,\ldots,m}$ is a finite family of flat finitely presented morphisms with $V_{j,i}$ affine, and the base change $\coprod_j T \times_{T_i} V_{j,i} \to T$ is surjective, then for some $i' \geq i$ the morphism $\coprod T_{i'} \times_{T_i} V_{j,i} \to T_{i'}$ is surjective. Denote $W_{i'} \subset T_{i'}$, resp. $W \subset T$ the image. Of course $W = T$ by assumption. Since the morphisms are flat and of finite presentation we see that W_i is a quasi-compact open of T_i, see Morphisms, Lemma 01UA. Moreover, $W = T \times_{T_i} W_i$ (formation of image commutes with base change). Hence by Limits, Lemma 01Z4 we conclude that $W_{i'} = T_{i'}$ for some large enough i' and we win.

For $\tau \in \{Zariski, étale, smooth, syntomic\}$ a standard τ-covering is a standard fppf covering. Hence the fully faithfulness of the functor holds. The only issue is to show that given a standard fppf covering $\mathcal{V}_i$ for some i such that $\mathcal{V}_i \times_{T_i} T$ is a standard τ-covering, then $\mathcal{V}_i \times_{T_i} T_{i'}$ is a standard τ-covering for all $i' \gg i$. This follows immediately from Limits, Lemmas 0EUU, 07RP, 0C0C, and 0C3L. □